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The Editor-in-Chief
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Dear Editor,

We are pleased to submit our original research manuscript entitled “**Pan-Andean explainable machine learning identifies priority glacial lakes for outburst flood early warning**” for consideration for publication in *Natural Hazards*.

Glacial lake outburst floods (GLOFs) are among the most destructive cryospheric hazards in the Andes, yet disaster-risk managers lack an objective, scalable method to decide which of the tens of thousands of Andean glacial lakes most urgently require inspection. Our study addresses this gap directly and, to our knowledge, presents the *first pan-Andean, explainable GLOF susceptibility model* validated against an independent historical catalogue. The main contributions are:

- A harmonized, multitemporal glacial-lake inventory of 69,643 lake-year observations (20,140 distinct lakes ≥ 0.05 km²) spanning fifteen study areas across four countries (Peru, Bolivia, Ecuador, and Chile, including Patagonia), built from free Sentinel-2 and Landsat imagery and open digital elevation models.
- A rigorously validated, *leakage-free* machine-learning framework evaluated under strict StratifiedGroup out-of-fold and Leave-One-Country-Out cross-validation, explicitly designed for the extreme class imbalance (1:629) and data scarcity ($n = 34$ GLOF lake-years from 32 distinct source lakes) inherent to historical GLOF catalogues. The best discriminator reaches AUC-ROC 0.826 (95% CI 0.729–0.916; permutation $p < 0.001$).
- SHAP-based physical interpretation that yields an honest, multivariate signature of GLOF-prone lakes—moderate size combined with compact shape, low moraine-dam freeboard, and proximity to active ice—rather than a single-variable proxy.
- A fully open-source, reproducible pipeline and an operational, isotonically calibrated (Brier Skill Score 0.24) prioritization framework: a compact top-2% watch-list of 403 lakes recovers 41% of confirmed GLOF-source lakes at 20.3× random enrichment (74.8× in the top 0.5%), directly usable by resource-constrained agencies such as Peru’s INAIGEM for evidence-based disaster risk reduction.

We believe the manuscript is well aligned with the scope of *Natural Hazards*, which covers flood and glacial hazards, risk assessment, and disaster risk reduction. Critically, the framework is designed for impact in *developing Andean nations* that bear the highest GLOF exposure yet have the fewest resources: it relies only on free Sentinel-2 imagery and open DEMs, runs without proprietary software, and turns these inputs into a ranked watch-list that lets resource-constrained agencies (e.g. Peru’s INAIGEM) triage scarce field-inspection budgets toward the lakes most likely to threaten

downstream communities. The work should thus be of broad interest to the hazard, cryosphere, remote-sensing, and disaster-risk-reduction communities, and provides a transferable template for other glacierized mountain regions.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submission and agree to its publication. The authors declare no competing interests. All input datasets are publicly available, and the full analysis pipeline, trained models, and code are openly released under the MIT licence.

We suggest the *Natural Hazards and Risk* subject area and would be glad to recommend potential reviewers upon request. Thank you for your consideration.

Sincerely,

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